SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I V Murale kaartik (192312567) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?
(5 Marks)

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis. (5 Marks)

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.
ii) Suggest two effective techniques to reduce overfitting in this model.
iii) Discuss how overfitting affects the model's generalization ability on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
(h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%
i) Identify the epoch at which the model starts overfitting.
ii) Explain why validation accuracy decreases despite training accuracy increasing.
iii) Suggest two methods to improve validation performance.

6. A machine learning model shows the following loss values during training: (5 Marks)

(a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
(b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3
(c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
(d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
(e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
(f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
(g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8
(h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1
i) At which epoch does the model achieve optimal generalization?
ii) Identify signs of overfitting from the data.
iii) Recommend two regularization techniques and explain their impact.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks)

a) Front identified as Front: 119
b) Front identifies as Back: 06
c) Back identified as Back: 120
d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a) Diseased identified as Diseased: 138
b) Diseased identified as Healthy: 12
c) Healthy identified as Healthy: 135
d) Healthy identified as Diseased: 15

Compute: **Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.**

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): (5 Marks)

- Defective identified as Defective: 2 30
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ITA0606.

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Sub:- Machine learning

CO1-At1

1. Given,

$P(A) = 1/4$
$P(B) = 1/2$
$P(C) = 1/4$

Blue probabilities:-

$P(Blue|A) = 4/12 = 1/3$
$P(Blue|B) = 7/15$
$P(Blue|C) = 2/15$

Using Bayes theorem:-

$$P(C|Blue) = \frac{P(C)P(Blue|C)}{P(A)P(Blue|A) + P(B)P(Blue|B) + P(C)P(Blue|C)}$$

$$P(C|Blue) = 1/4 \times 2/15 \Big/ 1/4 \times 1/3 + 1/2 \times 7/15 + 1/4 \times 2/15$$

$$\boxed{P(C|Blue) = 0.0851} \text{ (or) } \boxed{P(C|Blue) \approx 8.51\%}$$

2. Given,

$P(A) = 1/4$
$P(B) = 1/2$
$P(C) = 1/4$

Blue probabilities:-

- $P(Blue|A) = 3/10$
- $P(Blue|B) = 6/15 = 2/5$
- $P(Blue|C) = 2/15$

Using Baye's theorem:

$$P(C|Blue) = \frac{1/4 \times 2/15}{1/4 \times 3/10 + 1/2 \times 6/15 + 1/4}$$

$$\boxed{P(C|Blue) = 0.1081 \approx 10.81\%}$$

3. Total samples:-

$$1800 + 1800 = 3600$$

testing data = 20% :-

$$3600 \times 0.20 = 720$$

Given,

TP = 210

TN = 190

FP = 30

$\therefore$ FN = 720 - (210 + 190 + 30)

FN = 290

Metrics:-

$$\text{Accuracy} = \frac{TP+TN}{\text{Total}} = \frac{210+190}{720} = \boxed{55.56\%}$$

$$\text{Precision} = \frac{TP}{TP+FP} = 210/240 = \boxed{87.50\%}$$

$$\text{Sensitivity/Recall} = TP/TP+FN = 210/300 = \boxed{42\%}$$

$$\text{Specifity} = TN/TN+FP = 190/220 = \boxed{86.36\%}$$

$\therefore$ Precision and Specificity are high, but recall is only 42%, so the model may miss many malignant cases.

3/ Validation loss decreases until 'Epoch 3' & then starts increasing while training loss continues decreasing

1, Overfitting begins:-
Epoch 4

2, Two techniques:-
1, Early stopping - Stop training when validation loss starts increasing.
2, Dropout/Regularization - reduces dependence on specific training features.

3, Effect on generalization:-
Overfitting causes the model to learn training data too closely, reducing its performance on unseen data.

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Sol:- Validation accuracy increases upto Epoch 5 & then decreases.
i, Overfitting starts.
Epoch 5
ii, Why validation accuracy decreases:-
Training accuracy keeps increasing because the model learns the training data very closely while its ability to generalize to unseen validation data decreases.

iii, Two methods:-

1, Dropout

2, Early stopping

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Sol:- Validation loss:-

Epoch 1 → 2.7 Epoch 4 → 2.0

Epoch 2 → 2.3 Epoch 5 → 2.2

Epoch 3 → 2.1 Epoch 6 → 2.5

i, Optimal generalization:-

minimum validation loss occurs at Epoch 4

ii, Signs of Overfitting:-

- Training loss continued decreasing
- Validation loss starts increasing

iii, Two techniques.

- L2 regularization - penalizes large model weights
- Dropout - randomly disables neurons during training.

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Sol:- Given, TP = 119 FP = 5

FN = 6 Total = 250

TN = 120

Accuracy = $119+120/250 \times 100 = 95.60\%$

Precision = $119/119+5 \times 100 = 95.97\%$

Sensitivity = $119/119+6 \times 100 = 95.20\%$

Specificity = $120/120+5 \times 100 = 96.00\%$.

$$F1\,Score = \frac{2(119)}{2(119)+5+6} \times 100 = 95.58\%$$

Given,

TP = 138 TN = 135 total = 300

FN = 12 FP = 15

Accuracy = $138+135/300 \times 100 \Rightarrow 91\%$

Precision = $138/138+15 \times 100 \Rightarrow 90.20\%$

Sensitivity = $138/138+12 \times 100 \Rightarrow 92\%$.

Specificity = $135/135+15 \times 100 \Rightarrow 90\%$

F1 Score = $2(138)/2(138)+15+12 \times 100 \Rightarrow 91.09\%$.

Given, TP = 180 TN = 170 total = 400.

FN = 20 FP = 30

Accuracy = $180+170/400 \times 100 \Rightarrow 87.50\%$

Precision = $180/180+30 \times 100 \Rightarrow 85.71\%$

Sensitivity = $180/180+20 \times 100 \Rightarrow 90\%$

Specificity = $170/170+30 \times 100 = 85\%$.

F1 Score = $2(180)/2(180)+30+20 \times 100 = 97.80\%$.

Sol: Given,

TP = 230

FN = 20

TN = 225

FP = 25

Total = 500

Accuracy = $230+225/500 \times 100 = 91\%$

Precision = $230/230+25 \times 100 = 90.20\%$

Sensitivity = $230/230+20 \times 100 = 92\%$

Specificity = $225/225+25 \times 100 = 90\%$

F1 Score = $2(230)/2(230)+25+20 \times 100 = 91.09\%$.